University of Ottawa

Department of Mathematics and Statistics

MAT 1302C : Mathematical Methods II

Professor: Hadi Salmasian

Second Midterm Exam – Version A

November 21, 2013

Surname _____ First Name _____

Student # _____ DGD (Monday/Thursday) _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(d) Write your student number at the top of each page in the space provided.
(e) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(f) You are strongly recommended to write in **pen**, not pencil.
(g) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	3	3	4	5	5	3	23
Grade							

1. **[3 points]** Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 1 & 1 & -2 \\ 2 & 1 & -3 \end{bmatrix}.$$

2. **[3 points]** Let $A = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 2 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 1 \end{bmatrix}$. Solve the following matrix equation for X:

$$A(X^T + C) = BC$$

3. Let $A = \begin{bmatrix} 1 & 2 & 4 & 0 & 3 \\ 0 & 0 & -2 & 1 & -1 \\ 0 & 0 & 2 & -1 & 1 \end{bmatrix}$.

(a) **[3 points]** Find a basis for $\operatorname{Nul} A$.

(b) **[1 point]** What is $\operatorname{rank} A$? Justify your answer.

4. An economy consists of two sectors: the Construction sector and the Service sector. We know that:

- For each unit of output, Construction requires $\frac{1}{3}$ units from itself and $\frac{1}{2}$ units from Service.
- For each unit of output, Service requires $\frac{1}{6}$ units from Construction and $\frac{3}{4}$ units from iteself.

(a) [**1 point**] What is the consumption matrix C for this economy?

(b) [**2 points**] Find $I - C$ and $(I - C)^{-1}$.

(c) [**1 point**] Determine what intermediate demands are created if Construction wants to produce 12 units and Service wants to produce 24 units.

(d) [**1 point**] Find the production levels that will satisfy the final demand of 5 units from Construction and 10 units from Service.

5. Consider the matrix

$$A = \begin{bmatrix} -1 & -1 & 0 & 2 \\ 1 & 3 & 1 & -2 \\ 2 & 1 & 0 & -3 \\ 0 & 4 & -1 & 5 \end{bmatrix}.$$

(a) **[3 points]** Calculate $\det A$ by using the cofactor expansion method.

(b) **[2 points]** This part is independent of part (a).

Let P, Q, R be 3×3 matrices such that $\det(P) = -\frac{1}{2}$, $\det(Q) = 5$, and $\det(R) = -10$. Compute $\det(-\frac{1}{2}P^T Q^2 R^{-1} P^{-2})$.

6. Determine if the following sets are subspaces of $\mathbb{R}^3$ or not. Justify your answer.

(a) **[1 point]**

$$U = \left\{ \begin{bmatrix} a^2 + b^2 \\ b + c \\ 2c \end{bmatrix} \middle| \, a, \, b, \, c \in \mathbb{R} \right\}$$

(b) **[1 point]** $V = \left\{ \begin{bmatrix} a - 2b \\ -b + c \\ -2c \end{bmatrix} \middle| \, a, \, b, \, c \in \mathbb{R} \right\}$

(c) **[1 point]**

$$W = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \;\middle|\; x_1, x_2, x_3 \in \mathbb{R} \text{ and } \begin{bmatrix} 1 & 0 & -3 \\ 2 & 5 & -2 \\ 5 & 10 & -7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

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